0.1 Minimal Polynomial

Definition 0.1.1. Let V be a Vector Space over F, and let $T : V \to V$ be a linear operator. A map

$$\varphi_T : F[x] \to \mathcal{L}(V) : p(x) \mapsto p(T)$$

is called *evaluation* of T. It is a ring-homomorphism, and $\ker \varphi_T$ is principal ideal. Moreover,

$$F[x]/\ker \varphi_T \cong F[T]$$

by the first-isomorphism theorem.

Note: $\mathcal{L}(V) = \{T : V \to V \mid T \text{ is linear operator}\}$ has algebraic structures:

$(\mathcal{L}(V), +, \cdot)$ is Vector Space over F and $(\mathcal{L}(V), +, \circ)$ is a Ring with identity.

Generally, $(\mathcal{L}(V), +, \circ)$ is non-commutative, and <u>Not</u> domain. For instance, let $V = F^n$. Then, for each $1 \le j \le n$,

$$A_j : F^n \to F^n : (a_i)_{i=1}^n \mapsto (0, \ldots, a_j, 0, \ldots)$$

are not zero maps, but $E_1 \circ E_2 = \mathbf{0}$. And ask a toddler on the street, examples for Non-Commutative.
But, the subring

$$F[T] = \{p(T) \mid p(x) \in F[x]\} \subset \mathcal{L}(V)$$

is Commutative Ring with identity, still it is not a domain, in general. Moreover, as a Subspace,

$$F[T] = \mathsf{span}_F\{T^n \mid n = 0, 1, \ldots\}$$

If V is finite-dimensional space, then $\mathcal{L}(V)$ has a n^2 dimension, therefore

$$T^0, T^1, \ldots, T^{n^2}$$

is linearly dependent, this implies there exists a polynomial $p(x) \in F[x]$ such that $p(T) = 0$. In other words,

If V has a finite dimension, then for linear operator $T : V \to V$, $\ker \varphi_T$ is non-zero principal ideal.

In this case, the unique monic polynomial of $\ker \varphi_T$ is called the *minimal polynomial* for T.